Chapter 8
DemTect

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Abstract DemTect is a cognitive screening instrument, first published in 2000, which was designed to be sensitive to the early cognitive symptoms of dementia even in the stage of mild cognitive impairment. It covers a wide range of cognitive domains so that it is valid not only for patients with Alzheimer's disease but also for patients with other types of dementia. DemTect provides cutoff scores for dementia and for cognitive impairment typical of MCI. Much favored for cognitive screening purposes in Germany, English versions are also available.

Keywords DemTect • Cognitive screening • Dementia

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8.1 Introduction

The cognitive screening tool DemTect was first published in 2000 in a German version [1] and in 2004 in an English version [2]; also, a Polish [3, 4], a French [5], and some other versions are in use. The DemTect has attracted much attention since then and is not only recommended by German national guidelines [6] and authors reviewing cognitive screening tools (e.g. [7]) but also by international guidelines and recommendations to be used as a brief cognitive test for early detection of dementia [8] and mild cognitive impairment (MCI) [9, 10]. In a well-attended symposium on screening instruments at the conference of the German Society for Gerontopsychiatry and Psychotherapy (DGPPN, Deutsche Gesellschaft für Gerontopsychiatrie und psychotherapie) in 2005, the DemTect was elected as the favorite cognitive screening tool by the auditorium. In fact, the DemTect is the most used cognitive screening test in Germany next to the Mini-Mental State Examination (MMSE) [11].

8.2 Description of the Test

8.2.1 Subtests: Construction and Administration

The ambition of the DemTect construction was that it should (i) be sensitive to detect early cognitive symptoms of dementia even in the stage of MCI, (ii) have high specificity, (iii) cover a wide range of cognitive domains so that it is valid not only for patients with Alzheimer's disease (AD) for which assessment of learning and memory tests clearly is the most important issue but also for patients with other types of dementia, (iv) provide a total score that is independent of sociodemographic variables, and (v) provide cutoff scores for dementia but also a cutoff score that points to cognitive impairment rather belonging to the stage of MCI.

After some pilot work, five subtests were chosen for the DemTect (Table 8.1) that follow established test paradigms and which were able to fulfill the demands outlined above (for the rationale to select these subtests, see [2]):

1 and 5: Word list/delayed recall. A word list with ten words with immediate recall in two trials at the beginning of the test and a delayed recall at the end of the test (i.e. approximately 8 min later).
2: Number transcoding. A number transcoding task in which two Arabic numbers have to be transformed into verbal numerals and two verbal written numerals have to be transcoded into Arabic numbers (for typical errors in dementia patients as described in [12], see Fig. 8.1).
3: Verbal fluency. In the semantic verbal fluency task, the subjects have to name articles that can be bought in a supermarket within 1 min.
4: Digit span. In the digit span task, the subject has to repeat digits in reverse order to a maximum length of six.

With these subtests, the DemTect assesses short- and long-term verbal memory (word list), working memory (in the digit span task but also needed in the verbal

Table 8.1 Description of the DemTect subtests, its maximum raw scores, and its maximum transformed scores

DemTect subtest	Description	Max. raw score	Max. transformed score
Word list	Ten items have to be recalled in two trials; subjects are not informed of a delayed recall	20	3
Number transcoding	Two Arabic numbers have to be transformed into verbal numerals, and two verbal written numerals have to be transcoded into Arabic numbers	4	3
Verbal fluency	Within 1 min, the subjects have to name articles that can be bought in a supermarket (DemTect) or animals (DemTect B)	30	4
Digit span reverse	The subjects have to repeat digits in reverse order to a maximum length of six	6	3
Word list delayed recall	The ten items presented at the beginning of the test have to be recalled once more	10	5
Total transformed score			18

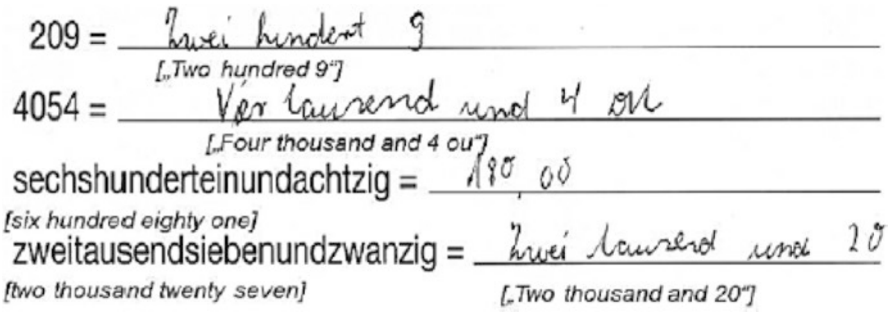

Fig. 8.1 Typical "shift errors," i.e. problems with shifting from one number code to the other (Arabic to number words or vice versa), and other errors in the number transcoding task in a patient with Alzheimer's disease

fluency task), executive functions (set shifting in the number transcoding task as well as cognitive flexibility in the verbal fluency task), and language (needed in all tasks but especially demanded in the verbal fluency task).

8.2.2 Scoring

The DemTect has a maximum transformed score of 18. The selection of this maximum score was random. For each subtest, transformation tables for two age groups (<60 years and ≥60 years) were provided for the first version of the DemTect. The maximum scores for each subtest range from 3 (word list, number transcoding, digit

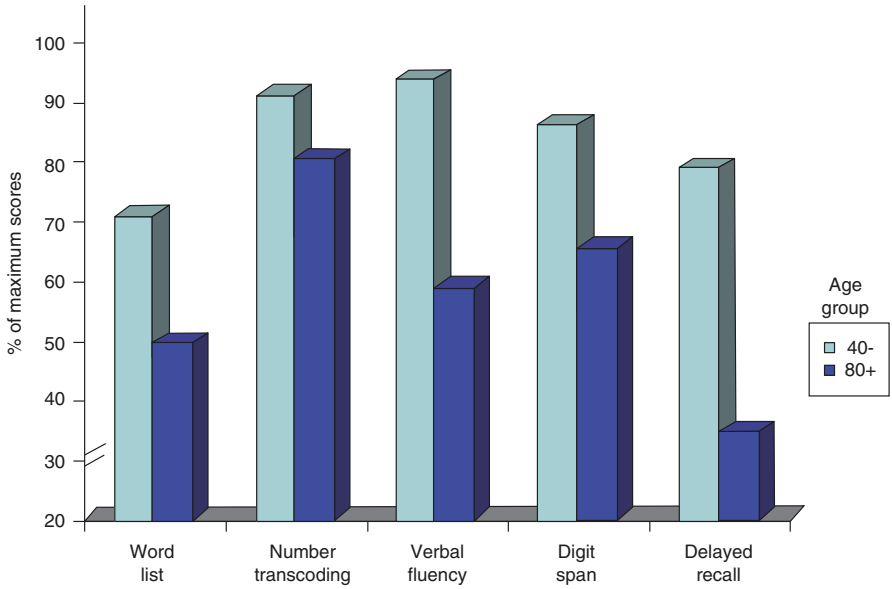

Fig. 8.2 Performance of the age groups "40–" (40 years and younger) and "80+" (80 years and older) (Modified according to [13]). Thirty words were taken as the maximum score for the verbal fluency task. The figure shows the age dependence of the different subtests

span) to 4 (verbal fluency) up to 5 (delayed recall). The decision on each maximum score was based on the subtests' different sensitivities and specificities in a population of healthy control subjects, AD patients, and MCI patients [1, 2]. The age correction was necessary due to significant age effects in the control groups in both normative studies. Furthermore, an education correction is provided in the English version [2]. Here, it was defined that one point is added to the transformed total score in subjects with only basic education (≤11 years).

After much feedback from clinicians that the DemTect is frequently used in elderly patients aged 80 years or above, but also in young patients of 40 years or younger (with a wide range of clinical states), further normative work was done by our own group [13] that has lead to norms for the age groups "40–" and "80+." With these scores, the total score of the DemTect is now independent of the factor age for adult patients from young adulthood until old age. The relevance of the age correction is demonstrated in Fig. 8.2.

8.2.3 Interpretation of the Total Transformed Score

From the transformed total DemTect scores, it can be decided whether performance of the subject can be interpreted as age adequate (13–18 points), or whether MCI

Table 8.2 Interpretation of DemTect scores

Transformed total score	Interpretation valid for DemTect and DemTect B scores
13–18 points	Cognitive abilities appropriate for the subjects' age
9–12 points	Mild cognitive impairment suspected
≤8 points	Dementia suspected

(9–12 points) or dementia must be suspected (≤8 points) (Table 8.2). Again, these scores were derived from the normative studies and show high sensitivity and specificity [1, 2].

It is important to emphasize that any interpretation from a screening tool must be preliminary; especially if a cognitive disorder is indicated, an elaborate neuropsychological examination is strongly recommended.

8.2.4 Administration Time

The administration time for the DemTect, including transformation of the raw scores and interpretation, is 8–10 min.

8.2.5 Avoiding Retest Effects with the Parallel Version of the DemTect: DemTect B

When patients are retested in follow-up examinations, explicit or implicit learning effects can occur when the same test versions are used. Thus, a parallel version of the DemTect, "DemTect B," was developed [14].

Parallel versions of the five original DemTect subtests were designed (modifications are indicated in Table 8.1). The equivalence of the new and original subtests was analyzed in 80 healthy control subjects. There were no significant differences between the corresponding subtests of the two test versions except for the semantic verbal fluency task (category "supermarket" in DemTect and category "animals" in DemTect B) (Fig. 8.3). Thus, different algorithms for transforming raw scores into transformed scores were calculated for this subtest. For all other subtests, the transformation tables of the original DemTect can be used. Using this procedure, there were no significant differences between the transformed scores of the DemTect and DemTect B, including the total scores (max. 18 points, mean score 15.9, SD 1.9 in DemTect versus 15.5, SD 2.4 in DemTect B). Thus, the interpretation of specific score ranges of the DemTect could be adopted for DemTect B, and the total DemTect B can be regarded as equivalent to the DemTect.

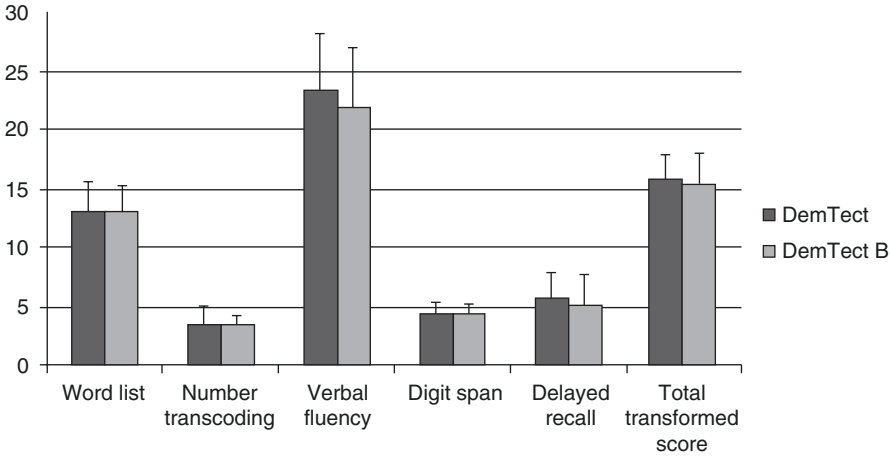

Fig. 8.3 Equivalence of performance in the parallel test versions DemTect and DemTect B in healthy control subjects

8.2.6 Psychometric Criteria

Besides the two normative studies for the German and English version of the DemTect [1, 2], some other studies have demonstrated a high sensitivity and specificity of the tool (overview in Table 8.3) [15]. The sensitivity across all studies ranges between 83 and 100 % for AD patients, 67 and 86 % for patients with MCI or mild cognitive disorder, and was 90 % for vascular dementia (VaD) patients; the specificity ranged between 90 and 100 % [1, 2, 16–18]. In a validation of the DemTect with 18-fluoro-2-deoxyglucose positron emission tomography (18-FDG-PET), the ROC analysis showed an area under the curve (AUC) of 0.78 with a cutoff score of ≤13 (95 % CI 0.62–0.94; $p=0.006$) [18].

The DemTect total transformed score is highly correlated with the MMSE (e.g. [2]; control group: $p<0.001$, $r=0.43$; AD group: $p<0.001$, $r=0.55$; MCI group: $p<0.01$, $r=0.31$). However, a regression analysis showed that although DemTect scores could be transformed into MMSE scores with the formula MMSE = 0.567 × DemTect score plus 19.997, DemTect scores only corresponded to MMSE scores higher than 20. This result reflects the fact that while the MMSE is a tool with which staging up to more severe stages of dementia is possible, the DemTect is a tool that is valuable for detecting and differentiating cognitive dysfunction when symptoms begin. Accordingly, the superiority of the DemTect compared to the MMSE regarding the sensitivity to assess early symptoms has been demonstrated [2, 16].

A good retest reliability with no significant differences in total transformed scores in 30 healthy controls which were tested two times with a time interval of 6 weeks (mean scores were 16.63 at t1 and 17.13 at t2) was shown [1].

Table 8.3 Sensitivity and specificity of the DemTect in studies with patients with dementia or mild cognitive impairment and healthy controls

Reference	Study samples	Sensitivity (sens.) and specificity (spec.)
Kessler et al. [1]	169 AD patients, 175 CG ($n=82<60$ yrs., $n=93\geq60$ yrs.)	AD versus CG ≥60 yrs.: sens.: 94 %, spec.: 90 %
Perneczky [16]	CG ($n=13$), AD patients ($n=13$), patients with mild cognitive disorder ($n=9$)	AD versus CG: sens.: 92 %, spec.: 100 %; mild cognitive disorder versus CG: sens.: 67 %, spec.: 92 %
Kalbe et al. [17]	AD patients ($n=36$), VaD patients ($n=28$), CG ($n=31$)	AD versus CG and VaD versus CG: sens. >90 %, spec.: >95 %
Kalbe et al. [2]	AD patients ($n=121$), MCI patients ($n=97$), CG ($n=145$)	AD versus CG: sens.: 100 %, spec.: 92 %; MCI versus CG: sens.: 86 %, spec.: 92 %
Scheurich et al. [18]	AD patients ($n=18$), MCI patients ($n=13$)	Sens. Compared to clinical diagnosis: AD: 83 %, MCI: 84.6 %; sens. compared to FDG-PET in all patients: 93 %

Modified from Kalbe et al. [15]
AD Alzheimer's disease, *CG* healthy control group, *yrs.* years, *VaD* vascular dementia, *MCI* mild cognitive impairment, *FDG-PET* 18-fluoro-2-deoxyglucose positron emission tomography

8.3 Neural Correlates of the DemTect Subtests

Neural correlates of the DemTect's five subtests regarding both gray matter brain atrophy and cerebral glucose metabolism were examined in 21 AD patients, 14 patients with frontotemporal lobar degeneration (FTLD), and 13 patients with subjective cognitive impairment (SCI) with structural magnetic resonance imaging (MRI) and F-18-fluorodeoxyglucose positron emission tomography (FDG-PET) by Woost et al. [19]. When all diagnostic groups were analyzed together, performance in the word list was positively correlated with glucose metabolism in the left temporal lobe. The number transcoding task was significantly related to glucose metabolism in a predominantly left lateralized frontotemporal network as well as a parietooccipital network including parts of the basal ganglia. Number transcoding was also associated with gray matter density in an extensive network including frontal, temporal, parietal and occipital areas. Working memory, tested with the digit span reverse, correlated with glucose metabolism in the left frontal cortex, the bilateral putamen, the head of caudate nucleus and the anterior insula. The only subtest for which no relationships with gray matter or glucose metabolism could be found was the supermarket task. Separate correlation analyses for the diagnostic groups partly verified or extended the correlates found for the overall sample analysis. The authors emphasize that their study serves as an external validation of the DemTect.

8.4 The DemTect in Clinical Practice and Scientific Contexts

The DemTect is a frequently used cognitive screening tool both in clinical practice and in scientific studies. Most of these studies and reports include patients with

dementia (e.g. [20–27]), or cognitive impairment [28]. However, the DemTect has also been used in patients with other neurological conditions [26], including patients with occipital, occipital-temporal and occipito-parietal infarction [29], patients with hypertension [30], implantable cardioverter-defibrillators [31], diabetes [32], primary hyperparathyroidism [33], possible osteoporosis [34], chronic lymphocytic leukemia [35], severe sepsis [36], and even in school children from 6 to 11 years to assess their cognitive functions [37]. Furthermore, the DemTect has been taken as an instrument to show effects of different kinds of interventions on cognitive functions, e.g. cognitive training [38–40] and cognitive and physical training in AD patients [41], herb extracts in elderly subjects with below-average cognitive performance [42], memantine in a female patient with alcohol-related dementia [43], deep brain stimulation in Parkinson's disease patients [44], provision of optical aid in patients with macular degeneration [45], and in patients with congestive heart failure receiving a biventricular defibrillator or implantable single or dual-chamber defibrillator [46]. The DemTect has also been used to demonstrate reduced quality of life in patients with cognitive impairment [47]. Finally, it was taken to test the criterion validity of the German version of the WHOQOL-OLD which is an instrument to assess the subjective quality of life in elderly people [48], to evaluate the MMSE in geriatric patients [49], and to evaluate psychometric criteria of a memory test to detect Alzheimer´s disease [50].

8.5 The "SIMARD: A Modification of the DemTect" – A Tool for the Identification of Cognitively Impaired Medically At-Risk Drivers

In 2011, a modification of the DemTect that aimed at identifying at-risk drivers was developed by a Canadian work group [51] who pointed out that physicians are well placed to identify medically at-risk drivers, but that there is a lack of valid screening tools that are easy to administer. Thus, the group carried out some research and validation work to develop such a brief screening tool for use in the primary care setting. The cohort comprised 146 consecutive referrals from community-based family physicians diagnosed with cognitive impairment or dementia and 35 community dwelling healthy controls who underwent an on-road evaluation with a subsequent "pass" or "fail" judgment. Among a set of neuropsychological tests, the best predictors for the on-road outcome was a combination of three DemTect subtests: the number conversion task, the supermarket task, and the repeat of the word list. With these three measures and with a modified scoring scheme, a further validation study with 123 individuals showed a sensitivity of the "SIMARD: A Modification of the DemTect" of 80 % and a specificity of 87 % for failing or passing in the on-road examination. Thus, the instrument can be regarded as a brief paper-and-pencil screening tool with a high degree of accuracy that can be used for immediate decisions on at-risk drivers in the clinical setting, although it has been critically discussed [52].

8.6 Conclusion

The DemTect, introduced in 2000 [1], is an easy-to-use cognitive screening tool that is valuable for the early detection of dementia and MCI. Important characteristics of the instrument are that it has an age correction and that its subtests are weighted according to their individual sensitivity and specificity – such as other screening tests developed by the same working group (review in [53]). It has attracted much attention both in clinical and scientific contexts. Other language versions exist (English, Polish, French, and others), a parallel test version, DemTect B, has been developed, and new normative data for subjects aged 40 years or younger and 80 years or older have been published. SIMARD, a modification of the DemTect, sensitive for the detection of elderly at-risk drivers, has been developed. Furthermore, the DemTect has been modified to permit assessment of cognitive functions in schoolchildren.

The sensitivity of the DemTect has been demonstrated in patients with AD, VaD, and MCI, but also various other diseases, and is superior to that of the MMSE. Its validity has also been shown with FDG-PET. Also, the DemTect has been included in studies with many conditions other than dementia or mild cognitive impairment as well as in studies that examine the effect of pharmacological and non-pharmacological interventions in various patient groups.

As for all cognitive screening instruments, it must be emphasized that these instruments can only serve as tools to detect patients suffering cognitive dysfunction. It represents the first step in a cascade of diagnostic procedures that, if a suspicion of decline has been verified by screening, include elaborate neuropsychological testing as well as extensive neurological and psychiatric examination. For this purpose though, screening tests are of crucial help. With its high sensitivity, easy administration and independency of sociodemographic factors, the DemTect fulfills all essential criteria for a cognitive screening instrument. It can be used by a wide range of professionals such as neuropsychologists, neurologists, or primary care physicians.

References

1. Kessler J, Calabrese P, Kalbe E, Berger F. DemTect: Ein neues Screening-Verfahren zu Unterstützung der Demenzdiagnostik. Psycho. 2000;26:343–7.
2. Kalbe E, Kessler J, Calabrese P, Smith R, Passmore AP, Brand M, Bullock R. DemTect: a new, sensitive cognitive screening test to support the diagnosis of mild cognitive impairment and early dementia. Int J Geriatr Psychiatry. 2004;19:136–43.
3. Wojtyńska R, Szczèniak D. DemTect ®-effective to asses MCI and dementia-validation study of the Polish language version. Aging Ment Health. 2016;20:510–6.
4. Calabrese P, Kalbe E, Kessler J, Fischer L, Smith B, Passmore P, Bullock R. A neuropsychological screening test to diagnose mild cognitive impairment and early dementia: DemTect. Psychogeriatria Polska. 2004;1:205–14.
5. Fischer-Altevogt L, Calabrese P, Kalbe E, et al. DemTect: a new diagnostic tool in the detection of dementia. Rev Geriatr. 2002;27:437–44.

6. DGN & DGPPN (Deutsche Gesellschaft für Neurologie & Deutsche Gesellschaft für Psychiatrie, Psychotherapie und Nervenheilkunde). S3-Leitlinie Demenzen. http://www.dgn.org/leitlinien/3176-leitlinie-diagnose-und-therapie-von-demenzen-2016. Accessed 27 Mar 2016.
7. Troyer AK. DemTect effective in screening for mild cognitive impairment and mild dementia. Evid Based Ment Health. 2004;7:70.
8. Feldman HH, Jacova C, Robillard A, Garcia A, Chow T, Borrie M, Schipper HM, Blair M, Kertesz A, Chertkow H. Diagnosis and treatment of dementia: 2. Diagnosis. CMAJ. 2008;178:825–36.
9. Chertkow H, Massoud F, Nasreddine Z, Belleville S, Joanette Y, Bocti C, Drolet V, Kirk J, Freedman M, Bergman H. Diagnosis and treatment of dementia: 3. Mild cognitive impairment and cognitive impairment without dementia. CMAJ. 2008;178:1273–85.
10. Jacova C, Kertesz A, Blair M, Fisk JD, Feldman HH. Neuropsychological testing and assessment for dementia. Alzheimers Dement. 2007;3:299–317.
11. Folstein MS, Folstein SE, McHugh PR. Mini-mental-state: a practical method for grading the cognitive state of patients for the clinician. J Psychiatr Res. 1975;12:189–98.
12. Kessler J, Kalbe E. Written numeral transcoding in patients with Alzheimer's disease. Cortex. 1996;32:755–61.
13. Kessler J, Fengler S, Kaesberg S, Müller K, Calabrese P, Ellwein T, Kalbe E. DemTect 40- und DemTect 80+: Neue Auswertungsroutinen für diese Altersgruppen. Fortschr Neurol Psychiatr. 2014;82:640–5.
14. Kessler J, Calabrese P, Kalbe E. DemTect-B: ein Äquivalenztest zum kognitiven Screening DemTect-A. Fortschr Neurol Psychiatr. 2010;78:532–5.
15. Kalbe E, Brand M, Kessler J, Calabrese P. Der DemTect in der klinischen Anwendung. Z Gerontopsychol Psychiatr. 2005;18:121–30.
16. Perneczky R. Die Eignung einfacher klinischer Tests für die Erkennung der leichten kognitiven Störung und der leichtgradigen Demenz. Akt Neurol. 2003;30:114–7.
17. Kalbe E, Kessler J, Smith R. DemTect: a new screeining instrument with very high sensitivity for vascular and Alzheimer dementia. In: Korczyn AD, editor. 2nd International congress on vascular dementia. Bologna: Monduzzi Editore; 2002. p. 129–33.
18. Scheurich A, Müller MJ, Siessmeier T, Bartenstein P, Schmidt LG, Fellgiebel A. Validating the DemTect with 18-fluoro-2-deoxy-glucose positron emission tomography as a sensitive neuropsychological screening test for early Alzheimer disease in patients of a memory clinic. Dement Geriatr Cogn Disord. 2005;20:271–7.
19. Woost T, Dukart J, Frisch S, Barthel H, Sabri O, Mueller K, Schroeter ML. Neural correlates of the DemTect in Alzheimer disease and frontotemporal lobar degeneration-A combined MRI & FDG-PET-study. NeuroImage Clin. 2013;2:746–58.
20. Rosengarten B, Paulsen S, Molnar S, Kaschel R, Gallhofer B, Kaps M. Activation-flow coupling differentiates between vascular and Alzheimer type of dementia. J Neurol Sci. 2007;257:149–54.
21. Rosengarten B, Paulsen S, Burr O, Kaps M. Neurovascular coupling in Alzheimer patients: effect of acetylcholinesterase inhibitors. Neurobiol Aging. 2009;30:1918–23.
22. Stoppe G, Buss K, Wolf S, Stiens G, Maeck L. Screening for dementia in very old age in primary care. Eur Psychiatry. 2010;25(Suppl1):589.
23. Schroeter ML, Vogt B, Frisch S, Becker G, Barthel H, Mueller K, Villringer A, Sabri O. Executive deficits are related to the inferior frontal junction in early dementia. Brain. 2012;135:201–15.
24. Eichler T, Thyrian JR, Hertel J, Köhler L, Wucherer C, Dreier A, Hoffmann W. Rates of formal diagnosis in people screened positive for dementia in primary care: results of the DelpHi-Trial. J Alzheimers Dis. 2014;42:451–8.
25. Eichler T, Wucherer D, Thyrian JR, Kilimann I, Hertel J, Michalowsky B, Teipel S, Hoffmann W. Antipsychotic drug treatment in ambulatory dementia care: prevalence and correlates. J Alzheimers Dis. 2015;43:1303–11.
26. Polidori MC, Stahl W, De Spirt S, Pientka L. Influence of vascular comorbidities on the antioxidant defense system in Alzheimer's disease. Dtsch Med Wochenschr. 2012;137:305–8.

27. Larner AJ. DemTect: 1-year experience of a neuropsychological screening test for dementia. Age Ageing. 2007;36:326–7.
28. Fib T, Fendrich K, van den Berg N, Meinke C, Hoffmann W. Prevalence and determinants for the intake of potentially inappropriate medication in patients with cognitive impairment: a retrospective analysis using the PRISCUS-criteria. 56th conference Deutsche Gesellschaft für Medizinische Informatik, Biometrie und Epidemiologie e.V. 2011. doi:10.3205/11gmds238.
29. Kraft P, Gadeholt O, Wieser MJ, Jennings J, Classen J. Lying obliquely: a clinical sign of cognitive impairment – cross sectional observational study. BMJ. 2009;339:b5273.
30. Blutdruckinstitut Göttingen e.V. Prävalenz kognitiver Funktionsstörungen bei Hypertonikern in der Praxis. DemTect Register. 2012. http://www.bdi-goe.de/forschung/demtect---register.php. Accessed 27 Mar 2016.
31. Prull MW, Unverricht S, Bittlinsky A, Sasko B, Wirdemann H, Gkiouras G, Butz T, Trappe HJ. Der ICD-Test führt zu einem zerebralen Schaden und verschlechtert die cognitive Funktion. Clin Res Cardiol. 2011;1000(Suppl1):V1532.
32. Rittmeier H. Therapierelevante kognitive Leistungsfähigkeit bei Patienten mit Typ 2 Diabetes. Master thesis, Konstanz; 2003.
33. Kätsch AK. Neuropsychiatric and cognitive changes after surgery for primary hyperparathyroidism. Dissertation, University of Düsseldorf, Düsseldorf; 2007.
34. Berkemeyer S, Schumacher J, Thiem U, Pientka L. Bone t-scores and functional status: a cross-sectional study on German elderly. PLoS One. 2009;4, e8216.
35. Goede V, Bahlo J, Chataline V, Eichhorst B, Dürig J, Stilgenbauer S, Kolb G, Honecker F, Wedding U, Hallek M. Evaluation of geriatric assessment in patients with chronic lymphocytic leukemia: results of CLL9 trial of the German CLL study group. Leuk Lymphoma. 2016;57:789–96.
36. Götz T, Günther A, Witte OW, Brunkhorst FM, Seidel G, Hamzei F. Long-term sequelae of severe sepsis: cognitive impairment and structural brain alterations-an MRI study (LossCog MRI). BMC Neurol. 2014;14:145.
37. Koch B, Hahn B. Der modifizierte DemTect als psychometrisches Screeninginstrument bei Grundschulkindern der Klassenstufen I bis IV. Neurol Rehabil. 2009;15:315–8.
38. Bohlken J. Kognitives Training in der Schwerpunktpraxis. Neurotransmitter. 2007;11:18–20.
39. Petrelli A, Kaesberg S, Barbe MT, Timmermann L, Fink GR, Kessler J, Kalbe E. Effects of cognitive training in Parkinson's disease: a randomized controlled trial. Parkinsonism Relat Disord. 2014;20:1196–202.
40. Petrelli A, Kaesberg S, Barbe MT, Timmermann L, Rosen JB, Fink GR, Kessler J, Kalbe E. Cognitive training in Parkinson's disease reduces cognitive decline in the long term. Eur J Neurol. 2015;22:640–7.
41. Drabben-Thiemann G, Hedwig D, Kenklies M, von Blomberg A, Marahrens G, Marahrens A, Hager K. Kognitive Leistungssteigerung bei Alzheimer-Erkrankten durch die Anwendung von Brain Gym. CO Med. 2005;3:118–22.
42. Berry NM, Robinson MJ, Bryan J, Buckley JD, Murphy KJ, Howe PRC. Acute effects of an Avena sativa herb extract on responses to the Stroop color-word test. J Altern Complement Med. 2011;17:635–7.
43. Bonnet U, Taazimi B, Borda T, Grabbe HD. Improvement of a woman's alcohol-related dementia via off-label memantine treatment: a 16-month clinical observation. Ann Pharmacother. 2014;48:1371–5.
44. Markser A, Maier F, Lewis CJ, Dembek TA, Pedrosa D, Eggers C, Timmermann L, Kalbe E, Fink GR, Burghaus L. Deep brain stimulation and cognitive decline in Parkinson's disease: the predictive value of electroencephalography. J Neurol. 2015;262:2275–84.
45. Mielke A, Wirkus K, Niebler E, Eschweiler G, Nguyen NX, Trauzettel-Klosinski S. The influence of visual rehabilitation on secondary depressive disorders due to age-related macular degeneration. A randomized controlled pilot study. Der Ophthalmologe: Zeitschrift der Deutschen Ophthalmologischen Gesellschaft. 2013;110:433–40.
46. Duncker D, Friedel K, König T, Schreyer H, Lüsebrink U, Duncker M, Oswald H, Klein G, Gardiwal A. Cardiac resynchronization therapy improves psycho-cognitive performance in patients with heart failure. Europace. 2015;17:1415–21.

47. Conrad I, Uhle C, Matschinger H, Kilian R, Riedel-Heller SG. Quality of life of individuals with mild cognitive impairment. Psychiatr Prax. 2015;42:152–7.
48. Conrad I, Matschinger H, Riedel-Heller SG, von Gottberg C, Kilian R. The psychometric properties of the German version of the WHOQOL-OLD in the German population aged 60 and older. Health Qual Life Outcomes. 2014;12:105.
49. Beyermann S, Trippe RH, Bähr AA, Püllen R. Mini-Mental State Examination in geriatrics: an evaluation of diagnostic quality. Z Gerontol Geriatr. 2013;46:740–7.
50. Szczesniak D, Wojtynska R, Rymaszewska J. Test Your Memory (TYM) as a screening instrument in clinical practice-the Polish validation study. Aging Ment Health. 2013;17:863–8.
51. Dobbs BM, Schopflocher D. The introduction of a new screening tool for the identification of cognitively impaired medically at-risk drivers: the SIMARD a modification of the DemTect. J Prim Care Community Health. 2010;1:119–27.
52. Bedard M, Weaver B, Man-Son-Hing M, Classen S, Porter M. Candrive Investigators. The SIMARD screening tool to identify unfit drivers: are we there now? J Prim Care Community Health. 2011;2:133–5.
53. Kalbe E, Calabrese P, Fengler S, Kessler J. DemTect, PANDA, EASY and MUSIC: cognitive screening tools with age correction and weighting of subtests according to their sensitivity and specificity. J Alzheimers Dis. 2013;34:813–34.